

Further/Extra Quizzes

Exercise

The directory /home/user contains the following files, which were derived from the output of the command `ls -l`.

```
-rw-r--r-- 1 scanzio root 0 mag 17 11:31 abXcx.txt
-rw-r--r-- 1 root scanzio 0 mag 17 11:38 a.py
-rw-r--r-- 1 scanzio root 0 mag 17 11:33 axcXab.txt
-rw-r-xr-- 1 scanzio root 0 mag 17 11:36 w.py
-rw-r--r-- 1 scanzio root 0 mag 17 11:33 Xcxab.txt
-rw-r--r-- 1 scanzio root 0 mag 17 11:33 x.py
-rw-r--r-- 1 scanzio root 3464324 mag 17 11:34 y.py
drwxr-xr-x 2 scanzio root 4096 mag 17 11:35 z.py
```

What are the files selected by the `find` command that follows:

```
find /home/user -type f \ ( -name "*.cpp" -o -name "*.py" -o -name "X?x*.txt" \) -
group root -s size -1M ! -perm 654
```

Choose one or more options:

1. ☐ z.py
2. ☐ y.py
3. ☐ x.py
4. ☐ Xcxab.txt
5. ☐ w.py
6. ☐ abXcx.txt
7. ☐ a.py
8. ☐ axcXab.txt

Correct answers: 2, 4

Exercise

Given the following bash command:

```
egrep -e "^[1-9]\ " -e "^1[0-9]\ |^2[0-1]\ " input.txt | egrep -e "[1-9][0-9]*\.[0-9]" | egrep "X$" | c ut -d " " -f 2,4
```

Report which lines are printed in output from the previous command, when it is cuted on the file `input.txt`, which is reported at the end of the question.

Content of the file `input.txt`

```
20 Stefano 12.5 X
22 Gabriele 13.9 Y
20 Giulia 7.4 X
15 Lodovica 17.3 A
16 Saverio 24.2 Y
21 Anna 12.3 X
2 Giorgio 14 X
```

Choose one or more options:

1. ☐ Lodovica 17.3 A
2. ☐ Giulia X
3. ☐ Gabriele 13.9 Y
4. ☐ Gabriele Y
5. ☐ Lodovica A
6. ☐ Anna X

7. ☐ Anna 12.3 X
8. ☐ Stefano X
9. ☐ Stefano 12.5 X

Correct answers: 2, 6, 8

Exercise

Given the execution in sequence of the following commands:

```
touch f1
ln -s f1 f2
ln f1 f3
ln f1 f4
echo "hello" > f3
rm f1
```

Indicate which (one or more) of the following outputs is (are) possible.

Choose one or more options:

1. ☐ After the command:
cat f4
no output is obtained
2. ☐ After the command:
ls -l
the following output is obtained:
lrwxrwxrwx 1 scanzio scanzio 2 ago 13 16:03 f2 -> f1
-rw-r--r-- 2 scanzio scanzio 6 ago 13 16:04 f3
-rw-r--r-- 2 scanzio scanzio 6 ago 13 16:04 f4
3. ☐ After the command:
cat f4
the following output is obtained:
hello
4. ☐ After the command:
cat f2
following output is obtained:
hello
5. ☐ After the command:
echo "pippo" > f2
ls -l
the following output is obtained:
-rw-r--r-- 1 scanzio scanzio 6 ago 13 16:19 f1
lrwxrwxrwx 1 scanzio scanzio 2 ago 13 16:03 f2 -> f1
-rw-r--r-- 2 scanzio scanzio 6 ago 13 16:04 f3
-rw-r--r-- 2 scanzio scanzio 6 ago 13 16:04 f4

Correct answers: 2, 3, 5

Exercise

Indicate which of the following statements related to the following sequence of shell commands are correct.

Note that wrong answers imply a penalty in the final score.

```
find / -name "a?b.c" -Execiseec cat \{} \; | tail -80 | head > /tmp/x
cat /tmp/x | sort -r | tr "e-g" "E-G"
```

Choose one or more options:

1. ☐ The command "sort -r" takes in input the content of the file /tmp/x, and it generates in output the same content but ordered in reverse alphabetical order
2. ☐ The file /tmp/x contains the first 10 rows of the files selected by the find command
3. ☐ The find command searches all the files which name starts with the letter "a", finishes with letters "b.c", and it is composed of any number of characters
4. ☐ The command tr "e-g" "E-G" substitutes the lowercase letters e, g with the corresponding uppercase letters (i.e., E, G)
5. ☐ The find command searches the files in the home directory
6. ☐ The find command searches all the files which name starts with the letter "a", finishes with letters "b.c", and it is composed of 5 characters

Correct answers: 1, 5, 6

Exercise

Assume that the following script is included in the file named script.sh:

```
#!/bin/bash echo "Number: $# " for x in $* do echo -n $x " " done
```

Also suppose it is invoked with the following command:

```
./script.sh "questa frase - $0 :" ha w parole e '$1' x
```

Indicate which of the following statements are correct. Note that wrong answers imply a penalty in the final score.

Choose one or more options:

1. ☐ L'output contains the string "\$0"
2. ☐ The output has a length of more than 2 lines
3. ☐ The output contains the string "\$1"
4. ☐ The output contains the string "Number: 6"
5. ☐ The output contains the string "Number: 7"
6. ☐ The output has a length of 2 lines
7. ☐ The value \$1 is substituted with 0

Correct answers: 3, 5, 6

Exercise

Given the following bash command:

```
egrep -e "^202101(2[7-8]|3[01])" input.txt | egrep -e "[1-9][0-9]*\.[0-9]" | egrep -e "13\.[127]" | cut -d " " -f 2-4
```

Report which lines are displayed when the previous command is Execiseecuted on the file input.txt reported at the end of the question.

Note that incorrect answers imply a penalty in the final score.

Content of the file input.txt

```
20210127 12.5 Stefano AXY
20210128 13.1 Giulia AXY
20210129 13.2 Gabriele AXY
20210130 13.3 Manuele AXY
20210131 13.7 Sara AXY
20210201 14.3 Enrico AXY
20210202 15.5 Pietro AXY
```

Choose one or more options:

1. ☐ 13.3 AXY

2. ☐ 13.7 Sara AXY
3. ☐ 13.1 AXY
4. ☐ 13.7 AXY
5. ☐ 13.3 Manuele AXY
6. ☐ 13.2 AXY
7. ☐ 13.1 Giulia AXY
8. ☐ 13.2 Gabriele AXY

Correct answers: 2, 7

Exercise

Using shell commands only, do the following:

1. Extract the 50-th row of a given file name.
2. save in a file named "list.txt" the names of all the files with extension ".h", which are contained in the root "/home/foo" directory tree and contain the string "define" or the string "include".
3. sort alphabetically in ascending order all lines in the file named "bar.txt", delete the duplicate lines, and pick only the second string of each line. Store the result in the file "/home/tmp/list.txt".

Indicate 3 separate commands using pipes or redirection operations.

Answer:

1.

```
cat "file.txt" | head -n 50 | tail -n 1
head -n 50 "file.txt" | tail -n 1
```

2.

```
find "/home/foo" -type f -name "*.h" -exec grep -l -E "(define)|(include)" \{} \; > "list.txt"
find "/home/foo" -type f -name "*.h" -exec grep -l -e "define" -e "include" \{} \; > "list.txt"
find "/home/foo" -type f -name "*.h" -exec grep -H -e "define" -e "include" \{} \; | cut -d ":" -f 1 > "list.txt"
grep -l -E "(define)|(include)" $(find "/home/foo" -name "*.h" -type f)
```

3.

```
cat "bar.txt" | sort | uniq | cut -d " " -f 2 > "/home/tmp/list.txt"
sort "bar.txt" | uniq | cut -d " " -f 2 > "/home/tmp/list.txt"
sort -u "bar.txt" | cut -d " " -f 2 > "/home/tmp/list.txt"
```

Exercise

In relation to the following shell command:

```
foo | bar 1> A 2> B
```

which of the following statements are correct:

Choose one or more options:

1. ☐ If you do not want to recover standard error, you can send it to /dev/null.
2. ☐ The foo command receives its input from the bar command and sends its standard output on file B and its standard error on A.
3. ☐ The bar command receives its input from the foo command and sends its standard output on file A and its standard error on file B.
4. ☐ The command would be identical if the symbol "|" were replaced with the character ";".
5. ☐ The command is incorrect and the symbol "|" should be replaced with the character ";".

6. ☐ The bar command receives its input from the foo command and sends its standard output on file B and its standard error on A.

Correct answers: 3

Exercise

Given the following shell command, indicate which of the following statements are correct. Note that incorrect answers imply a penalty in the final score.

```
find . -type f -regExercisetype posix-Exercisetended -regExercise  
".*/((XXXXXX(XX)*)|((BB|CCC){2,4}))([8-9][0-9]|1[0-4][0-9]|15[0-7])?.txt" -  
Exerciseec cat \{} \; | sort -k1,1nr| uniq
```

Choose one or more options:

1. ☐ The regular file ./BBCCCB.txt is selected by the find command.
2. ☐ The regular Exercisepression coded in the find command can match only files located in the current working directory.
3. ☐ The regular file ./BBCCCB148.txt is selected by the find command.
4. ☐ The regular file ./XXXXBBBB85.txt is selected by the find command.
5. ☐ For EACH selected file the find command performs a numerical sorting of its lines from the biggest to the smallest number, and if there are equal lines it deletes them.

Correct answers: 1, 3

Exercise

Given the following list of commands, indicate which of the following statements are correct after Exerciseecuting them. Note that incorrect answers imply a penalty in the final score.

```
touch f1  
ln f1 f2  
ln f1 f3  
ln -s f1 f4  
rm f1  
echo "hello" > f4  
echo "_world" >> f2
```

Choose one or more options:

1. ☐ File f1 was removed, and therefore it is not present in the directory.
2. ☐ Files f2 and f3 have the same content.
3. ☐ The dimension of file f4 is 2 bytes.
4. ☐ The number of hard links of file f2 is 2.
5. ☐ File f2 contains the string "hello_world".

Correct answers: 1, 2, 3, 4, 5

Exercise

Given the following Makefile, indicate which of the following statements are correct. Note that incorrect answers imply a penalty in the final score.

```
CC = gcc  
FLAGS = -Wall -g
```

```

LIB = -lm
.PHONY: clean distclean

mainVet.o: mainVet.c my.h
    $(CC) $(FLAGS) -c mainVet.c $(LIB)

inVet.o: inVet.c my.h
    $(CC) $(FLAGS) -c inVet.c $(LIB)

target: mainVet.o inVet.o
    $(CC) $(FLAGS) -o myExe mainVet.o inVet.o $(LIB)

distclean: clean1 clean2
    rm -f distclean

clean1: clean3
    rm -f clean1

clean2:
    rm -f clean2

clean3:
    rm -f clean3 rm -f clean3b

```

Choose one or more options:

1. ☐ Executing “make distclean” the sequence of commands that is executed is “rm -f distclean; rm -f clean2; rm -f clean1; rm -f clean3; rm -f clean3b”.
2. ☐ Executing the command "make", the default target that is named "target" is executed.
3. ☐ Executing two times in sequence “make target”, the command that is executed is “gcc -Wall -g -o myExe mainVet.o inVet.o -lm”.
4. ☐ Target listed after the keyword “.PHONY” are executed regardless the existence of a file with the same name that has been edited more recently than the dependencies

Correct answers: 3, 4

Exercise

Which of the following pieces of information are NOT stored inside the Process Control Block (PCB)?

Choose JUST ONE option:

1. ☐ Process state
2. ☐ List of open files
3. ☐ CPU registers
4. ☐ Signal handlers
5. ☐ Process Identifier (PID)
6. ☐ They are all stored within the PCB
7. ☐ Administrative data, for instance related to CPU usage
8. ☐ Program Counter

Correct Answers:

6

Exercise

Suppose that a process becomes "zombie".

Indicate which of the following statements are correct. Note that wrong answers imply a penalty in the final score.

Choose one or more options:

1. ☐ The PCB of the process was deleted
2. ☐ The process performed a wait
3. ☐ The process has not a parent process
4. ☐ The process is terminated
5. ☐ The process before terminating had the "init" process as parent process
6. ☐ The process was inherited by the process "init"
7. ☐ Its PCB will be deleted only after its parent performs a wait or a waitpid

Correct Answers:

4, 7

Exercise

Suppose a process becomes an "orphan."

Please indicate which of the following statements are correct. Note that incorrect answers imply a penalty in the final score

Choose one or more options:

1. ☐ The process is waiting that the parent performs a wait
2. ☐ The process becomes an orphan because it did not perform a wait
3. ☐ The process will become "zombie" at its termination
4. ☐ The process is inherited by the "init" process
5. ☐ The process will not become "zombie" at its termination because the "init" process will inherit it

Correct Answers:

4, 5

Ex

Analyze the following piece of code.

Please indicate which of the following statements are correct. Note that incorrect answers imply a penalty in the final score.

```
if (fork() == 0) {  
    /* first child */  
    if (fork() == 0) {  
        /* second child */  
        ...  
    } else {  
        exit (1);  
    }  
}  
wait ();
```

Choose one or more options:

1. ☐ The parent can suffer of deadlock
2. ☐ In case the first child has performed exit(1) , the second child will never become a zombie

3. ☐ To have a correct piece of code we must insert a second wait statement at the end
4. ☐ When the first child terminates, the second child is inherited by "init"
5. ☐ The parent waits the termination of the first child
6. ☐ The parent waits the termination of the first and the second child

Correct Answers:

2, 5

Exercise

Suppose you run the following segment of code:

Please indicate which of the following statements is correct.

```
if (fork()) {  
    sleep (10);  
    exit (1);  
} else {  
    exit (1);  
}
```

Choose JUST ONE option:

1. ☐ The child process will become an orphan.
2. ☐ The parent process will become zombie.
3. ☐ The child process will become zombie when the parent terminates.
4. ☐ The parent process will become an orphan.
5. ☐ The child process will become zombie.

Correct Answers:

5

Exercise

If the following program is run, how many characters 'P' will be displayed on standard output?

```
int main () {  
    int i;  
    i=0;  
    while (i<3 && fork()) {  
        fork ();  
        i++;  
    }  
    printf ("P");  
    return 1;  
}
```

Choose JUST ONE option:

1. ☐ 9
2. ☐ 16
3. ☐ 15
4. ☐ 8
5. ☐ 7

Correct Answers:

3

Exercise

Suppose a process does a waitpid. Please indicate which of the following statements are correct. Note that incorrect answers imply a penalty in the final score.

Choose one or more options:

1. ☐ The process will receive a SIGCHLD as soon as one of its children ends
2. ☐ The process can wait a maximum number of seconds
3. ☐ The process will exit from waitpid at the end of its first child
4. ☐ The process may get stuck on the waitpid even after a child has terminated

Correct Answers:

1, 4

Exercise

Describe what is an orphan process and then what is a zombie process.

Report two code segments: the first one generating an orphan process and the second one generating a zombie process.

Answer:

An orphan process is a child process in which the parent terminated before the child process. When this happens, orphan processes are "adopted" by a special os process (typically the init process). A zombie process is a child process that terminates before its parent called the system call wait().

```
// Orphan process
if(fork()){
    exit(0);
} else{
    // Orphan process
    sleep(1); // This orphan process will wait for a second so that we are sure
that the parent terminated
    exit(0);
}

// Zombie process
if(fork()){
    sleep(1); // Same as before but this time for the parent
    exit(0);
} else{
    //zombie process
    exit(0);
}
```

Exercise

Analyze the following segment of code. Indicate how many characters 'X' are displayed. Note that incorrect answers imply a penalty in the final score

```
#include <stdio.h>
#include <sys/types.h>
#include <unistd.h>
```

```

int main () {
    int i;
    i = 0;
    setbuf(stdout,0);
    while (i<=2 && fork()) {
        if (fork ()) {
            printf ("X");
        }
        i++;
    }
    return 1;
}

```

Choose JUST ONE option:

1. ☐ 14
2. ☐ 8
3. ☐ 6
4. ☐ 12
5. ☐ 7

Correct Answers:

5

Exercise

If you run the following program how many characters 'P' are displayed on standard output? Report a single integer value in your response.

```

int main () {
    int i;
    int pid;
    for (i=1; i<=2; i++) {
        pid = fork ();
        if (pid==0)
            fork();
    }
    printf("P");
    return 1;
}

```

Correct Answer:

9

Exercise

Indicate which of the following statements related to the fork() system call are correct. Note that incorrect answers imply a penalty in the final score.

Choose one or more options:

1. ☐ Parent and child share the code segment.
2. ☐ The fork() generates two new processes, to which are assigned two new pids.
3. ☐ Parent and child inherits the initial values of variables. (Meaning is: Parent and child inherit the values stored in the variables of the parent just before the fork())

4. ☐ The copy-on-write technique, after a `fork()`, permits to copy the initial values of variable in the parent and child processes.
5. ☐ Parent and child share the open file descriptors.

Correct Answer:

1, 3, 5

Exercise

Indicate which of the following statements related to process scheduling are correct. Note that incorrect answers imply a penalty in the final score.

Choose one or more options:

1. ☐ A process can move from the waiting state to the running state after an I/O operation or event completion.
2. ☐ The speedup that can be obtained from parallelization is limited by the fact that context switching operations reduce efficiency.
3. ☐ The process control block (PCB) contains all the information needed to manage the context switching, including a copy of the process stack and memory.
4. ☐ A process in the ready state does not use the CPU resource.
5. ☐ Different process queues are used to manage the access of the process to different resources when the process is in the waiting state.

Correct Answer:

2, 4, 5

Exercise

Indicate which of the following statements related to the system call `exec` are correct. Note that incorrect answers imply a penalty in the final score.

Choose one or more options:

1. ☐ In the case of correct execution of an `exec`, the instruction that follows the `exec` can be used to print a message of correct execution.
2. ☐ The `execl` used in the following way is correct: `execl("cp","cp","./file1","./file2",(char*)0);`
3. ☐ The `execv` has exactly two arguments, while the `execve` has exactly three arguments.
4. ☐ The `execlp` used in the following way is correct: `execlp("cp","abcdefgh","./file1","./file2",(char*)0);`
5. ☐ After an `exec`, the process identifier (pid) of the process is not changed.

Correct answer:

3, 4, 5

Exercise

- 1) Explain the effect of running the system call `exec` from a UNIX program.
- 2) Explain what differentiate the "l", "v", "p" and "e" versions.
- 3) Make an example of how the "v" and the "l" version can be used.
- 4) Why the system call `exec` should not return?

Answer:

- 1) The `exec` system call replaces the program associated with a process (code, data, etc.) with another one specified as a parameter, without replacing the PID (Process Identifier) of the initial process.
- 2) It basically changes the prototype of the function and what it receives as input:

l: list -> the function receives a list of arguments.

v: vector -> the function receives a vector of arguments.

p: path -> the function receives only the name of the file (without the path). The file is searched by browsing the list of directories stored in the environment variable with name PATH.

e: environment -> the function receives a vector containing a list of environment variables.

3) Example execv:

```
char *cmd[] = {"ls", "-l", (char *)0};  
execv("/bin/ls", cmd);
```

Example execl:

```
execl("/bin/ls", "ls", "-l", NULL);
```

4) The system call exec does not return to the caller program if executed with success. In the case of error, exec (in all its variants) returns -1.

Exercise

Suppose that the following program is executed passing the value 4 in the command line. Report the exact output generated by the program. Please, report the response in one line,, indicating all the messages and the values in output, which must be separated with only one space. Do not insert any other character in the response.

```
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
int main (int argc, char ** argv){  
    char str[100];  
    int i = atoi(argv[1]);  
    setbuf(stdout, 0);  
    printf("%d", i);  
    sleep(1);  
    if(i>0){  
        sprintf(str, "%s %d", argv[0], i-1);  
        system(str);  
    }  
    printf("End");  
}
```

Correct Answers:

4 3 2 1 0 End End End End End

Exercise

Suppose you are running the following program. Report the output it generates, remembering to respect exactly the output format produced.

Assume that the program is running on an operating system where a sleep of 1 second is long enough to complete all the other tasks currently running.

Please, write the answer on a single line, indicating the various output messages/values separated by a single space. Do not enter any other character in the response.

```
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
  
int main(){  
    int x;  
    x=0;  
    while (x<2 && fork()){  
        if (!fork())  
            execlp ("echo", "x++", "x", NULL);  
    }  
}
```

```

        x++;
        sleep (1);
        system("echo x+x");
    }
}

```

Correct Answer:

x x+x x x+x

Exercise

If the following program is run with a single parameter equal to the integer value 5, i.e.,

```
./pgrm 5
```

how many characters 'E' and how many characters 'e' will be displayed on standard output?

```

int main (int argc, char *argv[]) {
    char str[20];
    int n = atoi (argv[1]);
    setbuf (stdout, 0);
    if (n>0) {
        printf ("E");
        sprintf (str, "%d", n-1);
        execlp (argv[0], argv[0], str, NULL);
    }
    printf ("e");
    return 1;
}

```

Choose JUST ONE option:

1. ☐ 5 'E', 5 'e'.
2. ☐ 1 'E', 5 'e'.
3. ☐ 1 'E', 1 'e'.
4. ☐ 5 'E', 1 'e'.
5. ☐ 6 'E', 6 'e'.

Correct Answer:

4

Exercise

Analyze the following segment of code. Assume that in a few milliseconds all output operations are completed. Indicate which is the output generated by the program. Note that incorrect answers imply a penalty in the final score.

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
int main(){
    int i;
    for (i=0; i<4 && !fork(); i++){
        if (fork()) {
            sleep (1);
            system ("echo i+");
        } else
            execlp ("echo", "system", "i++", NULL);
    }
}

```

```

    }
}

```

Choose JUST ONE option:

1. ☐ i++ 0+ i++ 1+ i++ 2+ i++ 3+
2. ☐ 0++ 0+ 1++ 1+ 2++ 2+ 3++ 3+
3. ☐ 0++ i+ 1++ i+ 2++ i+ 3++ i+
4. ☐ i++ i+ i++ i+ i++ i+ i++ i+
5. ☐ i+ i++ i+ i++ i+ i++ i+ i++

Correct answer:

4

Exercise

Analyze the following segment of code. Assume that in a few milliseconds all output operations are completed. Indicate how many characters 'E' are displayed on standard output when the program is executed passing 3 as its first argument. Report a single integer value in your response.

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
int main (int argc, char *argv[]) {
    char str [50];
    int i;
    setbuf (stdout, 0);
    i = atoi (argv[1]);
    if (i<0)
        exit (0);
    if (fork () > 0) {
        if (fork () > 0) {
            sprintf (str, "echo -n E");
            system (str);
        }
        sprintf (str, "%d", i-2);
        printf ("E");
        execlp (argv[0], argv[0], str, NULL);
    }
    exit (0);
}

```

Correct Answer:

9

Exercise

Given the following segments of pseudo-code, which are executed concurrently by processes P1 and P2 with PID pid_P1 and pid_P2, respectively, indicate which of the following statements are TRUE.

```

P1
while (1) {
    ...
    kill (pid_P2, SIGUSR1);
    pause ();
}

```

```

        A();
    }

P2
while (1) {
    pause ();
    B();
    ...
    kill (pid_P1, SIGUSR2);
}

```

Choose one or more options:

1. ☐ It is possible to execute function A() more than one time (without the execution of any B() function in between)
2. ☐ Function A() can be executed before function B()
3. ☐ They are subject to deadlocks
4. ☐ They are subject to starvation
5. ☐ Function B() is certainly executed at least one time
6. ☐ Function B() is always executed before function A()

Correct Answers:

3, 4

Exercise

Describe the characteristics of the signals to send asynchronous information between processes. Implement the following program:

1. A program generates two processes P1 and P2 and it awaits for receiving signals from them. Every time it receives a message from P1 it displays on standard output the message "Signal received from P1". Analogously, every time it receives a message from process P2 it displays the message "Signal received from P2". If it receives 3 signals from the same process, it terminates processes P1 and P2 using the shell command "kill" and it terminates.

2. Process P1 and P2 run through an infinite cycle. Within the cycle, they await for a random time and then send a signal to the parent. P1 sends signal SIGUSR1; P2 sends signals SIGUSR2.

Answer:

```

#include<signal.h>
#include<sys/types.h>
#include<sys/wait.h>
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>

int last_sig = -1;
int last_last_sig = -1;
int finish = 0;

void sign_handler(int sig){
    if (sig==SIGUSR1)
        printf("Signal received from P1\n");
    else if (sig==SIGUSR2)
        printf("Signal received from P2\n");

    /* It receives 3 consecutive signals from the same process */
}

```

```

    if (sig == last_sig && last_sig == last_last_sig) {
        finish = 1;
    } else {
        last_last_sig = last_sig;
        last_sig = sig;
    }
}

int main() {
    char cmd[100];
    pid_t pid1, pid2;
    if ( (signal(SIGUSR1, sign_handler) == SIG_ERR) || (signal(SIGUSR2,
sign_handler ) == SIG_ERR) ) {
        printf("Error initializing signal handler");
        exit(-1);
    }
    pid1 = fork();
    if (!pid1) {
        /* P1 */
        while (1) {
            sleep( rand()%2 );
            kill(getppid(), SIGUSR1);
        }
    } else {
        pid2 = fork();
        if (!pid2) {
            /* P2 */
            while (1) {
                sleep( rand()%3 );
                kill(getppid(), SIGUSR2);
            }
        }
    }
    /* Parent */
    while (1) {
        pause();
        if (finish) {
            /* Kill P1 with a shell command */
            sprintf(cmd, "kill -9 %d", pid1);
            system(cmd);
            /* Kill P2 with a shell command */
            sprintf(cmd, "kill -9 %d", pid2);
            system(cmd);
            exit(0);
        }
    }
}

```

Exercise

Refer to the system call kill and to the kill shell command.

Indicate which of the following statements are correct. Note that wrong answers imply a penalty in the final score.

Choose one or more options:

1. ☐ The sent signal can be masked only in the case of the system call

2. ☐ Both the shell command and the system call can kill a process
3. ☐ The system call kills a process, the shell command does not
4. ☐ Both send a signal to a process
5. ☐ The shell command kills a process, the system call does not

Correct Answers:

2, 4

Exercise

Suppose to execute the following program with the value 4 passed on the command line. Report the output generated by the program. Suppose to run the program in an operating system where waiting of 1 second is sufficient to allow all other tasks that are running in that specific moment of time to terminate. Please, report the response on a single line, indicating the various messages and output values separated by a single space. Do not insert any other character into the answer. Format errors will be treated in the same way as other errors. This is an example of a correct answer: a 4 b++ etc.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main(){
    int i;
    for (i=0; i<4 && !fork(); i++){
        if (fork()) {
            sleep (1);
            system ("echo i+");
        }
        execlp ("echo", "system", "i++", NULL);
    }
}
```

Correct Answer:

i++ i+ i++

Exercise

Refer to the use of signals in a UNIX/Linux environment.

Indicate which of the following statements are correct. Note that incorrect answers imply a penalty in the final score.

Choose one or more options:

1. ☐ By using the system call signal() you can decide to ignore ANY type of signal
2. ☐ Inside a signal handler, only reentrant functions should be used
3. ☐ The reception of a signal by a process ALWAYS causes the termination of the process
4. ☐ The use of the pair of functions kill() and pause() or of the semaphore primitives sem_post() and sem_wait() are EQUIVALENT from the point of view of process synchronization
5. ☐ The reception of some types of signal (for instance SIGKILL) cannot be ignored
6. ☐ The execution of a signal handler after the reception of a signal by a process can lead to race conditions

Correct Answers

2, 5, 6

Exercise

Suppose a process executes the following instruction:

```
signal(SIGCHLD, SIG_IGN);
```

Indicate which ones among the following observations are correct (possibly more than one). Note that incorrect answers may imply a penalty on the final score.

Choose one or more options:

1. ☐ The process will handle signals of type SIGCHLD with the default function SIG_IGN (which is defined as a macro in signal.h).
2. ☐ The process will not have to execute a wait or a waitpid.
3. ☐ The process will behave in a standard way when a SIGCHLD is received.
4. ☐ The process will ignore signals of type SIGCHLD.
5. ☐ If the process executes a wait, it will return an error code.
6. ☐ The child process can become zombie.

Correct Answers

1, 2, 4, 5

Exercise

Indicate which of the following statements related to signals are correct. Note that incorrect answers imply a penalty in the final score.

Choose one or more options:

1. ☐ If a process receives a signal and then it executes a pause(), it will remain blocked on it (as long as it does not receive another signal).
2. ☐ After the reception of a signal the process has three possibilities: accept the default behavior, ignore it (not for all signals), or catch the signal by defining a signal handler.
3. ☐ Signal can be transferred only between processes with the same parent.
4. ☐ The use of the kill() always leads to the termination of the process to which it is addressed.
5. ☐ The system call signal() can be used to send a signal.

Correct Answers

1, 2

Exercise

Given three processes P1, P2 and P3 whose code is reported in the following and whose pids are pid_P1, pid_P2, and pid_P3, respectively. Indicate which of the following outputs are correct. Note that incorrect answers imply a penalty in the final score.

```
P1
kill(pid_P2, SIG...);
pause();
printf("A");
P2
other_code();
pause();
printf("B");
kill(pid_P1, SIG...);
kill(pid_P3, SIG...);
P3
```

```
other_code();
pause();
printf("C");
```

Choose one or more options:

1. ☐ No output.
2. ☐ B
3. ☐ BC
4. ☐ BA
5. ☐ BCA
6. ☐ BAC

Correct Answers

1, 2, 3, 4, 5, 6

Exercise

Consider the following statement: "If one thread terminates, all other threads in the process will necessarily terminate." Indicate in which circumstances among the following ones the previous statement is correct. Note that incorrect answers imply a penalty in the final score

Choose one or more options:

1. ☐ When the thread performs a return from its start function.
2. ☐ When the thread performs and exit.
3. ☐ When the thread performs a return from the main.
4. ☐ When the thread receives a pthread_cancel from another thread.
5. ☐ When the thread performs a pthread_exit.

Correct Answers: 2. 3

Exercise)

Explain why:

1. Two processes can or cannot share a global variable.
2. Two threads can or cannot share a global variable.

Answer:

1. During the generation of a new process, for instance by means of the system call fork(), the address space is duplicated and therefore disjoint.
2. Threads share the same address space, as a consequence a write operation on a global variable by means of a thread has effect on all the other threads

Exercise

In a relatively unloaded multiprocessor system (where response times are fast), what does the following fragment of code produce on standard output?

```
int i=0; pthread_t thread, thread2;
void *t1(void *a){
    pthread_detach (pthread_self ());
    printf ("%d", ++i);
    return NULL;
}
```

```

void *t2(void *a){
    sleep(1);
    printf ("%d", ++i);
    return NULL;
}

int main() {
    if(fork())
        pthread_create (&thread, NULL, t1, NULL);
    sleep(1);
    if(fork())
        pthread_create (&thread2, NULL, t2, NULL);
    printf("A\n");
}

```

Choose JUST ONE option:

1. ☐ AA
2. ☐ 1AAAA
3. ☐ 12AAAA
4. ☐ AAAA
5. ☐ 1AA
6. ☐ 12AA

Correct Answers: 2

Exercise

A multi-threaded program consists of several threads. Thread A executes an `exit()` before terminating, thread B a `pthread_exit()`, and thread C a `return`.

Indicate which of the following statements are correct. Note that wrong answers imply a penalty in the final score.

Choose one or more options:

1. ☐ In order not to terminate the other threads, the `exit()` must be performed only by the initial function of the thread (e.g., `main()`)
2. ☐ All other threads end with thread B
3. ☐ All other threads end with thread C
4. ☐ In order to terminate, thread C must perform the `return` from its initial function of the thread (e.g., `main()`)
5. ☐ All other threads end with thread A

Correct Answers: 4, 5

Exercise

Suppose to execute the following program with the value 4 passed on the command line. Report the output generated by the program. Please, report the response on a single line, indicating the various messages and output values separated by a single space. Do not insert any other character into the answer. Format errors will be treated in the same way as other errors. This is an example of a correct answer: 2 5 0 3

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

```

```

#include <pthread.h>

pthread_t thread;
int i;

void *t1 (void *a){
    int *p;
    p = (int *) a;
    i = *p;
    pthread_detach (pthread_self ());
    printf ("%d ", i);
    i--;
    if (i>0)
        pthread_create (&thread, NULL, t1, (void *) &i);
    return NULL;
}

int main(int argc, char **argv) {
    i = atoi (argv[1]);
    if (fork())
        pthread_create (&thread, NULL, t1, (void *) &i);
    sleep (1);
    printf ("%d ", -i);
}

```

Answer:

4 3 2 1 0 -4

Exercise

Analyze the following piece of code.

Please indicate which of the following statements are correct. Note that incorrect answers imply a penalty in the final score.

```

void *thread_main(void *p) {
    int x, *y;
    y = (int *)p;
    x = *y;
    x += x;
    *p = x;
    return NULL;
}

int main() {
    int data = 1;
    pthread_t one, two;
    pthread_create(&one, NULL, thread_main, &data);
    pthread_create(&two, NULL, thread_main, &data);
    pthread_join(one, NULL);
    pthread_join(two, NULL);
    printf("%d\n", data);
    return 0;
}

```

Choose one or more options:

1. ☐ The code contains a race condition

2. ☐ The code does not contain a race condition
3. ☐ The value 1 can be printed
4. ☐ The value 0 can be printed
5. ☐ The value 4 can be printed
6. ☐ The value 2 can be printed

Correct Answers: 1, 5, 6

Exercise

Analyze the following segment of code.

Please indicate which of the following statements are correct. Note that incorrect answers imply a penalty in the final score.

```
void* myfunc(void* ptr) {
    int *tidP, data;
    data = (int *) ptr;
    printf("%d ", data);
    return NULL;
}

int main() {
    int i, x;
    void *retval;
    pthread_t tid[10];
    for (i=0; i<10; i++) {
        pthread_create(&tid[i], NULL, myfunc, (void *) &i);
    }
    for (x=0; x<10; x++) {
        pthread_join (tid[x], &retval);
    }
    pthread_exit(NULL);
}
```

Choose one or more options:

1. ☐ The code can display the sequence: 1 2 3 4 5 6 7 8 9 11
2. ☐ The code contains a race condition
3. ☐ The code can display the sequence: 1 2 3 4
4. ☐ The code does not contain a race condition
5. ☐ The code can display the sequence: 1 2 3 4 5 6 7 8 9 10
6. ☐ The code can display the sequence: 1 7 8 8 8 8 8 8 8 10
7. ☐ The code can display the sequence: 0 1 2 3 4 5 6 7 8 9

Correct Answers: 2, 5, 6, 7

Exercise

Suppose a thread executes the following instruction:

```
pthread_detach (pthread_self ());
```

Which among the following observations are correct (possibly more than one). Note that incorrect answers may imply a penalty on the final score.

Choose one or more options:

1. ☐ The thread that executed this instruction CANNOT perform a pthread_join.

2. ☐ The thread that created this thread CANNOT perform a `pthread_join` (with tid related to the thread that executed the instruction `pthread_detach`).
3. ☐ The thread that executed this instruction CANNOT perform a `pthread_create`.
4. ☐ The status information will be lost at the termination of the thread.
5. ☐ The thread that executed this instruction CANNOT perform a `pthread_exit`.

Correct Answers: 2, 4

Exercise

Analyze the following segment of code.

Indicate which is the output generated by the program. Note that incorrect answers imply a penalty in the final score.

```
#include <stdio.h>
#include <pthread.h>

void *t2 ();
int n = 5;
int m = 3;
void *t1 () {
    pthread_t thread;
    if (n>0) {
        printf ("%d ", n--);
        pthread_create (&thread, NULL, t2, NULL);
    }
    pthread_join (thread, NULL);
    pthread_exit (NULL);
}

void *t2 () {
    pthread_t thread;
    if (m>0) {
        printf ("%d ", m--);
        pthread_create (&thread, NULL, t1, NULL);
    }
    pthread_join (thread, NULL);
    pthread_exit (NULL);
}

int main (int argc, char *argv[]) {
    pthread_t thread;
    setbuf (stdout, 0);
    pthread_create (&thread, NULL, t1, &n);
    pthread_join (thread, NULL);
    return 1;
}
```

Choose JUST ONE option:

1. ☐ 3 2 1 5 4 3 2
2. ☐ 5 3 4 2 3
3. ☐ 5 3 4 2 3 1 2 0 1
4. ☐ 5 3 4 2 3 1 2
5. ☐ 5 4 3 2 3 2 1

Correct Answers: 4

Exercise

Indicate which of the following statements related to threads are correct. Note that incorrect answers imply a penalty in the final score.

Choose one or more options:

1. ☐ To communicate threads can use global variables.
2. ☐ The function `pthread_join()` waits the termination of a specific thread whose identifier is passed as a parameter to the function.
3. ☐ Each thread with the exception of the thread with `tid=0` has a parent.
4. ☐ The parallel execution of an algorithm by means of more than one thread leads IN ANY CASE to a reduction on its execution time.
5. ☐ The function `pthread_exit()` executed by one thread leads to the termination of the only thread that executes it.
6. ☐ The instruction `return` executed by one thread leads IN ANY CASE to the termination of the whole process.

Correct Answers: 1, 2, 5

Exercise

In the following code snippet, the main program creates N threads that execute the function `tF`. The purpose of the program is to execute the threads passing to each one an integer value that uniquely identifies the thread, i.e., an integer number ranging from 0 to N-1.

Indicate which of the following observations are correct. Note that incorrect answers imply a penalty in the final score.

```
#define N 10
void *tF (void *par) {
    int *tidP, tid;
    ...
    tidP = (int *) par;
    tid = *tidP;
    ...
    pthread_exit (NULL);
}
int main () {
    pthread_t t[N];
    int rc, i;
    for (i=0; i<N; i++) {
        rc = pthread_create (&t[i], NULL, tF, (void *) &i);
        if (rc) {
            fprintf (stderr, "Error.\n");
            exit (1);
        }
    }
    pthread_exit (NULL);
}
```

Choose one or more options:

1. ☐ The variable `tid` can contain the value 10 for one thread.

- 2. ☐ The variable tid contains a different integer value for each thread.
- 3. ☐ The variable tid can contain the value 10 for all the threads.
- 4. ☐ The variable tid can contain the value 7 for all the threads.
- 5. ☐ A copy of the variable is passed to each thread.

Correct Answers: 1, 3